

Laser Cutting Ring and Pin Gear Mechanism Using Fusion 360

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1. Introduction

This project focuses on the design and fabrication of a ring and pin gear mechanism using Fusion 360 and laser cutting technology. The mechanism demonstrates the working principle of internal gear systems where a smaller pin gear rotates inside a larger ring gear. The project combines CAD modeling, digital fabrication, and mechanical design concepts to create a functional educational model.

2. Objectives of the Project

The main objective was to design a precise and functional ring and pin gear mechanism suitable for laser cutting. Another aim was to improve knowledge of Fusion 360 tools and understand the relationship between gear ratios and motion transfer.

3. Design Software

Fusion 360 was used to create the sketches, define the dimensions, and generate the final 3D model. The software provided accurate tools for designing the gear teeth, circular holes, and support structure.

4. Ring Gear Design

The ring gear was designed with internal teeth to allow smooth meshing with the pin gear. Circular holes were added around the outer ring to reduce material usage and improve the appearance of the model.

5. Pin Gear Design

The pin gear was designed carefully to fit inside the ring gear with proper tooth alignment. The number of teeth and diameter were selected to ensure efficient motion transfer and smooth rotation.

6. Material Selection

Acrylic and plywood are common materials used for laser cutting because they are lightweight and easy to machine. The thickness of the sheet material was selected based on the required strength and durability of the mechanism.

7. Laser Cutting Process

The final Fusion 360 sketches were exported as DXF files and imported into laser cutting software. The cutting parameters such as speed, power, and focus were adjusted to achieve clean and accurate cuts.

8. Assembly Procedure

After the cutting process, the gears were assembled using support pins and spacers. Careful alignment ensured that the gears rotated smoothly without excessive friction or misalignment.

9. Testing and Results

The mechanism was tested manually to observe the movement of the gears. The ring and pin gear system operated successfully, demonstrating smooth motion and proper engagement between the teeth.

10. Advantages of the Model

The project is economical, lightweight, and suitable for educational demonstrations. It helps students understand gear systems, rotational motion, CAD design, and laser cutting technology in a practical way.

11. Conclusion

The ring and pin gear mechanism project successfully demonstrated the use of Fusion 360 for mechanical design and laser cutting fabrication. The completed model achieved proper gear interaction and stable movement. This project improved practical skills in CAD modeling, digital manufacturing, and mechanical engineering concepts.

Model Images



